

UI/UX Screen Specification — Reports & Analytics Module

Reuses RGD v2.0, Appendix A-N, and prior UI/UX Screen Specifications by reference only. FR-40 to FR-42, BR-RPT-001 to BR-RPT-005 cited by ID. Reports & Analytics is a pure consumption layer (Appendix-I Section 14) with no entities of its own; domain-level report screens (Attendance, Examination, Fee, HR, Library, Transport) consolidate and cross-reference the identical specifications already defined in their respective modules (SCR-ATT-009, SCR-EXM-014, SCR-FEE-013, SCR-HR-010, SCR-LIB-013, SCR-TRN-010), not restated as new logic. Report Scheduler references a schedule field not modeled in Appendix-G/H — flagged inline (*) as a Schema Extension item.

#	Screen ID	Screen Name
1	SCR-RPT-001	Reports Dashboard
2	SCR-RPT-002	Student Reports
3	SCR-RPT-003	Attendance Reports
4	SCR-RPT-004	Examination Reports
5	SCR-RPT-005	Fee Reports
6	SCR-RPT-006	HR Reports
7	SCR-RPT-007	Library Reports
8	SCR-RPT-008	Transport Reports
9	SCR-RPT-009	Report Scheduler
10	SCR-RPT-010	Report Export History

SCR-RPT-001 — Reports Dashboard

Purpose	Role-scoped landing page aggregating widget-level KPIs across all modules, per FR-40.
User Roles	Management, Academic Head, Finance Team, IT Admin, HR
Navigation Path	Reports > Dashboard
Page Header	"Reports Dashboard"
Breadcrumb	Home > Reports > Dashboard
Form Sections	Role-Scoped Widget Grid, Quick Links to Domain Reports
Field Display Order	Enrollment/Attendance Summary (widget) -> Fee Collection Summary (widget) -> Exam Performance Summary (widget) -> Quick Links
Conditional Visibility Rules	Widget set differs per role (Appendix-I Section 8.3 Role-to-Module Access Matrix) -- e.g., Finance Team sees Fee widgets prominently, Academic Head sees Attendance/Exam widgets.
UI Behaviour	Widgets refresh on load; no inline editing (BR-RPT-002, read-only enforcement).
Buttons & Actions	Go to Student Reports, Go to Fee Reports, Go to Exam Reports, Customize Dashboard
Button Placement	Widget-level action links; "Customize" top-right.
Toolbar Actions	N/A
Table Columns	N/A (widget-based)
Default Visible Columns	N/A
Hidden Columns	N/A
Row Actions	N/A
Filters	Academic Session
Default Sorting	N/A
Default Pagination	N/A
Bulk Actions	N/A
Export Formats	N/A
Print Behaviour	N/A
Empty State	"No report data available for this session."
Loading State	Skeleton widgets while data loads.
Popup/Dialog Details	N/A
Audit Information	N/A (summary view).
Responsive Behaviour	Widgets stack vertically on narrow viewports (NFR-MOB-001).
Role Permissions	All listed roles: Read (widget set scoped per role).
Related Business Rules	BR-RPT-002 (read-only enforcement), FR-40

Success/Error Condition	Message
Load Error	"Unable to load dashboard. Please retry."

SCR-RPT-002 — Student Reports

Purpose	Generate enrollment, demographic, and category-distribution reports drawn from the Student entity, per FR-41.
User Roles	Academic Head, Admin Staff, Management, IT Admin (read-only)
Navigation Path	Reports > Student Reports
Page Header	"Student Reports"
Breadcrumb	Home > Reports > Student Reports
Form Sections	Report Filters, Report Output
Field Display Order	Report Type -> Class/Section -> Academic Session -> Generate
Conditional Visibility Rules	"Category Distribution" report type reveals an RTE-specific breakdown sub-section (BR-ADM-003 cross-reference).
UI Behaviour	Loading Indicator: shown during generation; large reports process as a background job (NFR-RPTP-001).
Buttons & Actions	Generate Report, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Class/Section, Total Students, Category Breakdown
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Drill Down (to Student List, SIS module)
Filters	Academic Session, Class, Section, Category
Default Sorting	Class, ascending
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Print-ready summary layout.
Empty State	"No data available for the selected filters."
Loading State	"Generating report..." with background-job notice for large ranges.
Popup/Dialog Details	N/A
Audit Information	Report generation logged (generated-by, timestamp, filters) per Appendix-J Section 15.
Responsive Behaviour	Table becomes horizontally scrollable on narrow viewports.
Role Permissions	Academic Head/Admin Staff/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-RPT-002 (read-only reporting), FR-41

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Type	Dropdown	Yes	Enrollment Summary/Category Distribution/Section Strength	Enrollment Summary	—	—	Editable

Success/Error Condition	Message
Success	"Report generated."
Queued	"Large report queued -- you will be notified when ready."

SCR-RPT-003 — Attendance Reports

Purpose	Consolidated entry point to attendance summary/trend/defaulters reports; full specification reused from Attendance Module SCR-ATT-009.
User Roles	Academic Head, Management, IT Admin (read-only)
Navigation Path	Reports > Attendance Reports
Page Header	"Attendance Reports"

Breadcrumb	Home > Reports > Attendance Reports
Form Sections	Report Filters, Report Output (identical to SCR-ATT-009)
Field Display Order	Report Type -> Class/Section -> Date Range -> Generate
Conditional Visibility Rules	Identical to SCR-ATT-009 -- "Defaulter List" reveals a Minimum % Threshold field.
UI Behaviour	Loading Indicator: shown during generation; background job for large ranges (NFR-RPTP-001).
Buttons & Actions	Generate Report, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Class/Section, Students, Present %, Absent %, Below-Threshold Count
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Drill Down (to Class Attendance Register, SCR-ATT-007)
Filters	Class, Section, Date Range, Report Type
Default Sorting	Present %, ascending
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Print-ready summary layout.
Empty State	"No data available for the selected filters."
Loading State	"Generating report..." with background-job notice for large ranges.
Popup/Dialog Details	N/A
Audit Information	Report generation logged per Appendix-J Section 15.
Responsive Behaviour	Table becomes horizontally scrollable on narrow viewports.
Role Permissions	Academic Head/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-ATT-006, BR-RPT-002; consolidates SCR-ATT-009 (Attendance Module)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Type	Dropdown	Yes	Daily Summary/Trend/Defaulter List	Daily Summary	—	—	Editable

Success/Error Condition	Message
Success	"Report generated."
Queued	"Large report queued."

SCR-RPT-004 — Examination Reports

Purpose	Consolidated entry point to result trend/performance analytics; full specification reused from Examination Module SCR-EXM-014.
User Roles	Academic Head, Management, IT Admin (read-only)
Navigation Path	Reports > Examination Reports
Page Header	"Examination Reports"
Breadcrumb	Home > Reports > Examination Reports
Form Sections	Analytics Filters, Trend Charts, Summary Table (identical to SCR-EXM-014)
Field Display Order	Report Scope -> Academic Session Range -> Class/Subject Filter -> Generate
Conditional Visibility Rules	Chart view replaces table view when Report Scope = Session-over-Session Trend.
UI Behaviour	Loading Indicator: shown during generation; background job for large multi-session queries (NFR-RPTP-001).
Buttons & Actions	Generate, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	Chart/Table View Toggle
Table Columns	Class/Subject, Average %, Pass %, Above-Threshold Count
Default Visible Columns	All visible

Hidden Columns	None
Row Actions	Drill Down (to Student Result History, SCR-EXM-013)
Filters	Academic Session Range, Class, Subject
Default Sorting	Average %, descending
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Chart and table export to print-ready PDF.
Empty State	"No data available for the selected filters."
Loading State	"Generating analytics..." with background-job notice.
Popup/Dialog Details	N/A
Audit Information	Report generation logged per Appendix-J Section 15.
Responsive Behaviour	Charts resize to viewport; table horizontally scrollable on narrow viewports.
Role Permissions	Academic Head/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-RPT-002, BR-RPT-004 (provisional-data labelling); consolidates SCR-EXM-014 (Examination Module)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Scope	Dropdown	Yes	Class-wise/Subject-wise/Session-over-Session Trend	Class-wise	—	—	Editable

Success/Error Condition	Message
Success	"Analytics generated."
Queued	"Large analytics query queued."

SCR-RPT-005 — Fee Reports

Purpose	Consolidated entry point to collection summary, outstanding-dues, RTE-reimbursement, and GST reports; full specification reused from Fee Module SCR-FEE-013.
User Roles	Finance Team, Management, IT Admin (read-only)
Navigation Path	Reports > Fee Reports
Page Header	"Fee Reports"
Breadcrumb	Home > Reports > Fee Reports
Form Sections	Report Filters, Report Output (identical to SCR-FEE-013)
Field Display Order	Report Type -> Academic Session -> Class -> Date Range -> Generate
Conditional Visibility Rules	"RTE Reimbursement" report type reveals a "Reimbursement Period" field.
UI Behaviour	Loading Indicator: shown during generation; background job for large reports (NFR-RPTP-001).
Buttons & Actions	Generate Report, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Class/Category, Total Billed, Total Collected, Outstanding, Collection %
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Drill Down (to Due & Defaulter List, SCR-FEE-012)
Filters	Academic Session, Class, Date Range, Report Type
Default Sorting	Collection %, ascending
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Print-ready summary layout.
Empty State	"No data available for the selected filters."
Loading State	"Generating report..." with background-job notice.

Popup/Dialog Details	N/A
Audit Information	Report generation logged per Appendix-J Section 15.
Responsive Behaviour	Table becomes horizontally scrollable on narrow viewports.
Role Permissions	Finance Team/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-RPT-002, FR-26a/FR-26b; consolidates SCR-FEE-013 (Fee Module)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Type	Dropdown	Yes	Collection Summary/Outstanding Dues/RTE Reimbursement/GST Summary	Collection Summary	—	—	Editable

Success/Error Condition	Message
Success	"Report generated."
Queued	"Large report queued."

SCR-RPT-006 — HR Reports

Purpose	Consolidated entry point to payroll summary, statutory-filing, and department-cost reports; full specification reused from HR & Payroll Module SCR-HR-010.
User Roles	HR, Finance Team, Management, IT Admin (read-only)
Navigation Path	Reports > HR Reports
Page Header	"HR Reports"
Breadcrumb	Home > Reports > HR Reports
Form Sections	Report Filters, Report Output (identical to SCR-HR-010)
Field Display Order	Report Type -> Pay Period Range -> Department -> Generate
Conditional Visibility Rules	"Statutory Filing" report type reveals a "Filing Period" field distinct from the general Pay Period Range.
UI Behaviour	Loading Indicator: shown during generation; background job for large reports (NFR-RPT-001).
Buttons & Actions	Generate Report, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Department, Employee Count, Gross Total, Deductions Total, Net Total
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Drill Down (to Payroll Processing, SCR-HR-007)
Filters	Pay Period Range, Department, Report Type
Default Sorting	Department, alphabetical
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Print-ready summary layout.
Empty State	"No data available for the selected filters."
Loading State	"Generating report..." with background-job notice.
Popup/Dialog Details	N/A
Audit Information	Report generation logged per Appendix-J Section 15.
Responsive Behaviour	Table becomes horizontally scrollable on narrow viewports.
Role Permissions	HR/Finance Team/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-RPT-002, BR-HR-005; consolidates SCR-HR-010 (HR & Payroll Module)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Type	Dropdown	Yes	Payroll Summary/Statutory Filing (PF/ESI/PT)/Department Cost	Payroll Summary	—	—	Editable

Success/Error Condition	Message
Success	"Report generated."
Queued	"Large report queued."

SCR-RPT-007 — Library Reports

Purpose	Consolidated entry point to circulation, overdue, and inventory reports; full specification reused from Library Module SCR-LIB-013.
User Roles	Librarian, Management, IT Admin (read-only)
Navigation Path	Reports > Library Reports
Page Header	"Library Reports"
Breadcrumb	Home > Reports > Library Reports
Form Sections	Report Filters, Report Output (identical to SCR-LIB-013)
Field Display Order	Report Type -> Date Range -> Category Filter -> Generate
Conditional Visibility Rules	"Category Filter" hidden unless Report Type = Inventory.
UI Behaviour	Loading Indicator: shown during generation; background job for large reports (NFR-RPT-001).
Buttons & Actions	Generate Report, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Title/Category, Issues, Returns, Overdue Count, Fine Collected
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Drill Down (to Book Master, Library Module)
Filters	Date Range, Category, Borrower Type
Default Sorting	Issues, descending
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Print-ready summary layout.
Empty State	"No data available for the selected filters."
Loading State	"Generating report..." with background-job notice.
Popup/Dialog Details	N/A
Audit Information	Report generation logged per Appendix-J Section 15.
Responsive Behaviour	Table becomes horizontally scrollable on narrow viewports.
Role Permissions	Librarian/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-RPT-002; consolidates SCR-LIB-013 (Library Module)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Type	Dropdown	Yes	Circulation Summary/Overdue List/Inventory/Fine Collection	Circulation Summary	—	—	Editable

Success/Error Condition	Message
Success	"Report generated."
Queued	"Large report queued."

SCR-RPT-008 — Transport Reports

Purpose	Consolidated entry point to route occupancy, vehicle utilization, and maintenance-due reports; full specification reused from Transport Module SCR-TRN-010.
User Roles	Transport Coordinator, Management, IT Admin (read-only)
Navigation Path	Reports > Transport Reports
Page Header	"Transport Reports"

Breadcrumb	Home > Reports > Transport Reports
Form Sections	Report Filters, Report Output (identical to SCR-TRN-010)
Field Display Order	Report Type -> Route -> Academic Session -> Generate
Conditional Visibility Rules	"Maintenance Due" report type reveals a "Due Within (days)" field.
UI Behaviour	Loading Indicator: shown during generation; background job for large reports (NFR-RPT-001).
Buttons & Actions	Generate Report, Export, Schedule Recurring Report
Button Placement	Top-right action bar.
Toolbar Actions	N/A
Table Columns	Route/Vehicle, Capacity, Allocated, Utilization %
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Drill Down (to Route Master/Vehicle Master, Transport Module)
Filters	Academic Session, Route, Report Type
Default Sorting	Utilization %, descending
Default Pagination	50 rows/page
Bulk Actions	N/A
Export Formats	PDF, Excel, CSV
Print Behaviour	Print-ready summary layout.
Empty State	"No data available for the selected filters."
Loading State	"Generating report..." with background-job notice.
Popup/Dialog Details	N/A
Audit Information	Report generation logged per Appendix-J Section 15.
Responsive Behaviour	Table becomes horizontally scrollable on narrow viewports.
Role Permissions	Transport Coordinator/Management: Read, Export. IT Admin: Read.
Related Business Rules	BR-RPT-002; consolidates SCR-TRN-010 (Transport Module) -- Vehicle Maintenance data itself remains flagged in Transport Module (SCR-TRN-009)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Type	Dropdown	Yes	Route Occupancy/Vehicle Utilization/Fee Collection/Maintenance Due	Route Occupancy	—	—	Editable

Success/Error Condition	Message
Success	"Report generated."
Queued	"Large report queued."

SCR-RPT-009 — Report Scheduler

Purpose	Configure a recurring generation/dispatch schedule for any report available in this module.
User Roles	Academic Head, Finance Team, HR, Management, IT Admin
Navigation Path	Reports > Report Scheduler
Page Header	"Report Scheduler"
Breadcrumb	Home > Reports > Scheduler
Form Sections	Scheduled Report List, Add Schedule
Field Display Order	Report Type -> Recurrence -> Recipients -> Next Run Date
Conditional Visibility Rules	"Recipients" list filtered to users holding Read/Export permission for the selected Report Type (Appendix-I Section 8.3).
UI Behaviour	Confirm Dialog: shown on schedule creation. Unsaved Changes Warning: yes.
Buttons & Actions	Add Schedule, Edit, Pause, Delete
Button Placement	"Add" top-right; others inline per row.
Toolbar Actions	N/A
Table Columns	Report Type, Recurrence, Recipients, Next Run Date, Status
Default Visible Columns	All visible

Hidden Columns	None
Row Actions	Edit, Pause/Resume, Delete
Filters	Report Type, Status
Default Sorting	Next Run Date, ascending
Default Pagination	25 rows/page
Bulk Actions	N/A
Export Formats	N/A (operational queue screen)
Print Behaviour	N/A
Empty State	"No scheduled reports."
Loading State	Loading indicator on save.
Popup/Dialog Details	Delete-schedule confirm dialog.
Audit Information	Schedule creation/edit/delete events logged with actor, timestamp.
Responsive Behaviour	Table condenses on narrow viewports.
Role Permissions	Academic Head/Finance Team/HR/Management: Create, Read, Update, Delete (own schedules). IT Admin: full authority.
Related Business Rules	* Schema Extension – Client/Product Decision Required. (no ScheduledReport entity in Appendix-G/H; report generation is currently on-demand only)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Report Type	Dropdown	Yes*	Any report available in this module	—	Select report	—	Editable
Recurrence	Dropdown	Yes*	Daily/Weekly/Monthly	Weekly	—	—	Editable
Recipients	Multi-select	Yes*	Valid User(s) with Read access to the selected report	—	Select recipients	—	Editable

Success/Error Condition	Message
Success	"Report schedule saved. Next run: {date}."
Validation Error	"At least one recipient is required."

SCR-RPT-010 — Report Export History

Purpose	View the log of all report generation and export actions across the module, per Appendix-J Section 15.
User Roles	IT Admin, Management (own exports)
Navigation Path	Reports > Export History
Page Header	"Report Export History"
Breadcrumb	Home > Reports > Export History
Form Sections	Export Log List
Field Display Order	Date Range -> Report Type Filter -> Export Log List
Conditional Visibility Rules	Result set scoped to own export actions for non-IT-Admin roles.
UI Behaviour	List refreshes on filter change; no inline editing (read-only historical log).
Buttons & Actions	Re-download, Export Log
Button Placement	"Re-download" inline per row; "Export Log" top-right.
Toolbar Actions	N/A
Table Columns	Report Type, Generated By, Filters Used, Format, Generated Date
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Re-download
Filters	Report Type, Date Range, Format
Default Sorting	Generated Date, descending
Default Pagination	30 rows/page
Bulk Actions	N/A
Export Formats	Excel, CSV (log export)
Print Behaviour	N/A

Empty State	"No export history found."
Loading State	Loading indicator on filter change.
Popup/Dialog Details	N/A
Audit Information	This screen is itself a read view of the report-generation audit trail (Appendix-J Section 15); no separate audit layer applies to it.
Responsive Behaviour	Table becomes stacked cards on narrow viewports.
Role Permissions	IT Admin: Read (all). Management/Academic Head/Finance Team/HR: Read (own exports only).
Related Business Rules	Appendix-J Section 15 (report generation logging, reused unchanged)

Success/Error Condition	Message
Load Error	"Unable to load export history."